

Assignment 1: Part B (Big Data Analysis)

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Revised Questions

After receiving feedback for the previous assignment, as well as coincidentally revising my research direction with my supervisor for Research Project, I spent some time revising my research direction for this project. The revised questions are more specific while tackling a problem with significantly less complexity.

Primary Question:

How accurately can a CNN trained on AI4MARS classify 320x320 patches as traversable or non-traversable, and how does co-training with the terrestrial Freiburg Forest dataset improve that accuracy?

Backup Question:

If the Freiburg Forest domain transfer proves to be unhelpful, can advanced augmentation boost performance alone?

Revised Datasets

- AI4MARS: A Dataset for Terrain-Aware Autonomous Driving on Mars
 - NASA Open Data Portal
- Freiburg Forest Multi-Modal/Spectral Annotated dataset
 - Autonomous Intelligent Systems DeepScene

Data Snapshot

Metric	AI4MARS (Mars)	Freiburg Forest (Earth)
Labelled images	16064 train / 322 test	230 train / 136 test
Native Size	1024x1024	Assorted
Center Crop	320x320	320x320
Pixel-wise Classes	4 (soil, bedrock, sand, other)	7 (road, grass, vegetation, tree, sky, obstacle, void)
Binary Rule	Soil & Bedrock = Safe	Road = Safe

Key Observations

- Freiburg Forest's annotated subset consists of only 366 images
 - Augmentation & Class-Balancing will likely be required
- Mars images feature low-saturation, grey hues
 - Earth imagery features variety of green and brown hues

Feature Exploration

Colour & Intensity:

The hue distribution histogram (Figure 2) shows Mars pixels being concentrated near hue = 0 (low-saturation grayscale imagery). Meanwhile, you can see the Earth pixels cover a wider band of hues with a peak around 15-25 (brown soil/road) and a long low distribution of higher hues coming from the vegetation and sky.

Embedding/Cluster Insight:

- PCA of Mars test features colour-coded by the Mars+Earth predictions (Figure 4) reveal that the decision boundary aligns with the PC 1 texture gradient
 - Shows us that texture is a discriminating feature and the Mars+Earth model's boundary can be interpreted
- t-SNE on the 5-D feature vector (Figure 3) separates Mars (red) from Earth (blue) but still shows internal clustering
 - This is evidence that some feature cues, like rock texture, are shared

Simple Classification Baseline

Train Set	Precision (Safe)	Recall (Safe)	Precision (Hazardous)	Recall (Hazardous)	Balanced Accuracy
Mars Only	0.994	0.647	0.322	0.979	0.813
Mars + Earth	0.995	0.673	0.338	0.979	0.826

Observation:

Even with a tiny Earth set consisting of 366 images, the balanced accuracy improves by 0.013. The increase comes from less misclassifications (Figure 5). This supports our theory that terrestrial data transfers to Martian terrain.

Findings & Next Steps

- Colour and texture features differ slightly but still overlap
 - Common hazard labels suggest positive transfer potential
- Earth co-training improves performance
 - Small but consistent boost in balanced accuracy
- Amount of terrestrial data is limited compared to Martian data
 - Might benefit from augmentation and class-balanced sampling

Refined Plan

1. Train light CNN as first deep baseline
2. Explore other potential classification architectures
3. Experiment with self-supervised pre-training on unlabelled Freiburg Forest raw dataset

References

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Appendix

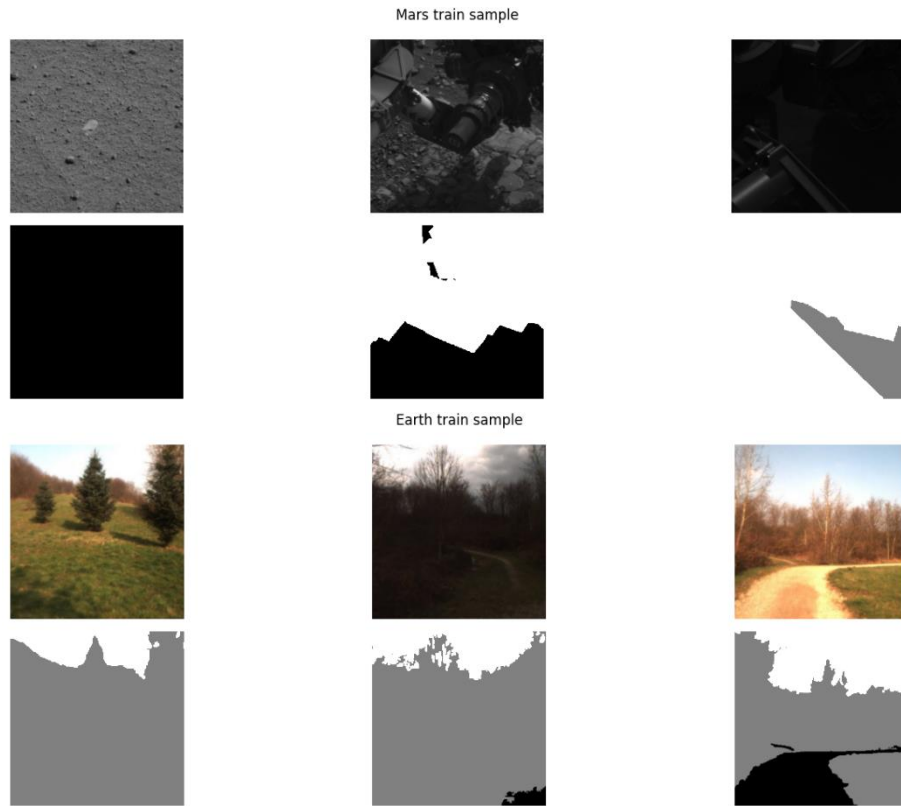


Figure 1 - Random Mars vs Earth image / mask montage (sanity check)

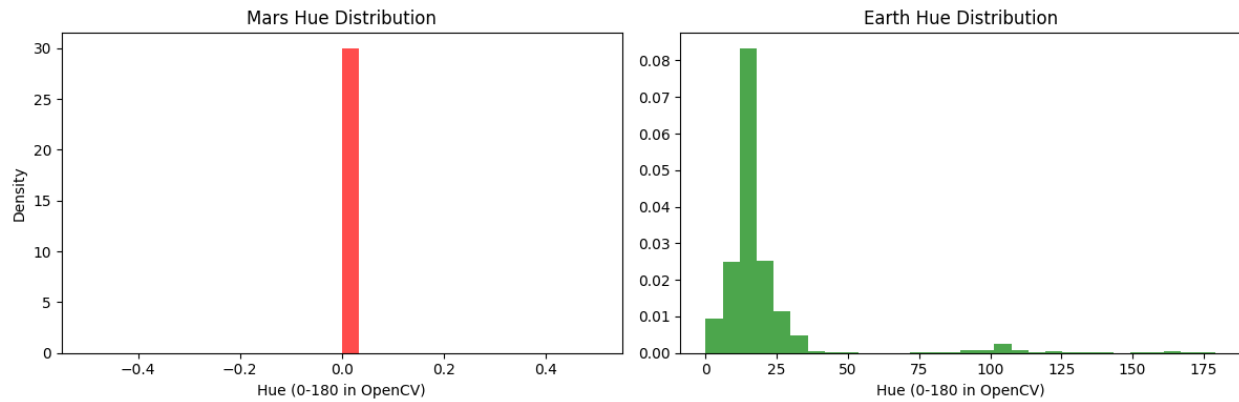


Figure 2 - Separate hue histograms for Mars and Earth

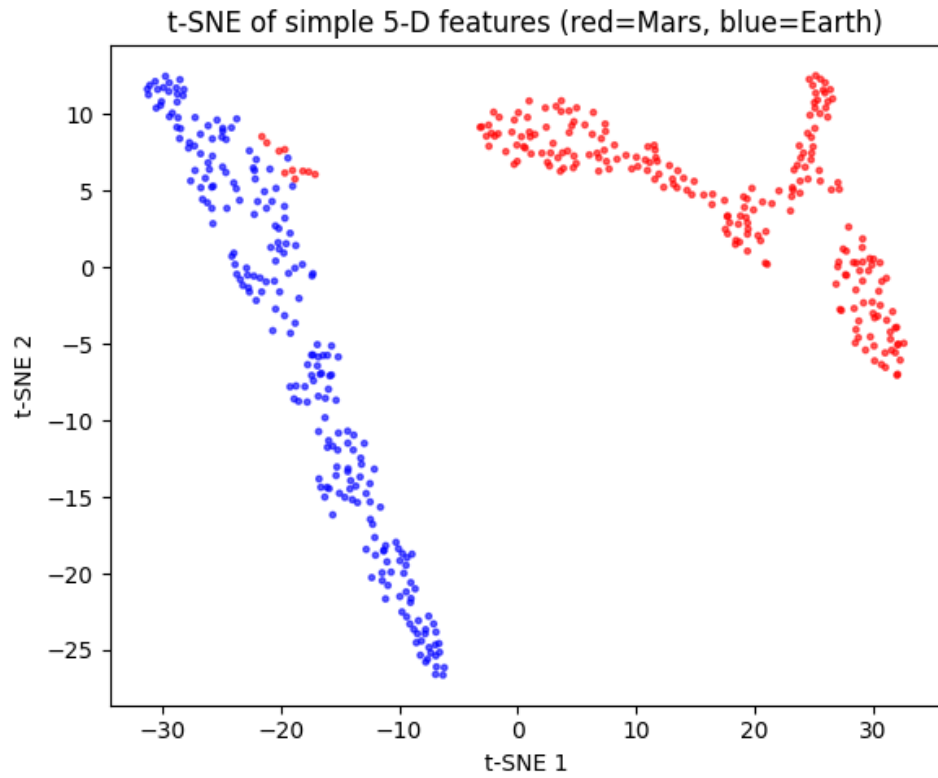


Figure 3 - t-SNE of 5-D features

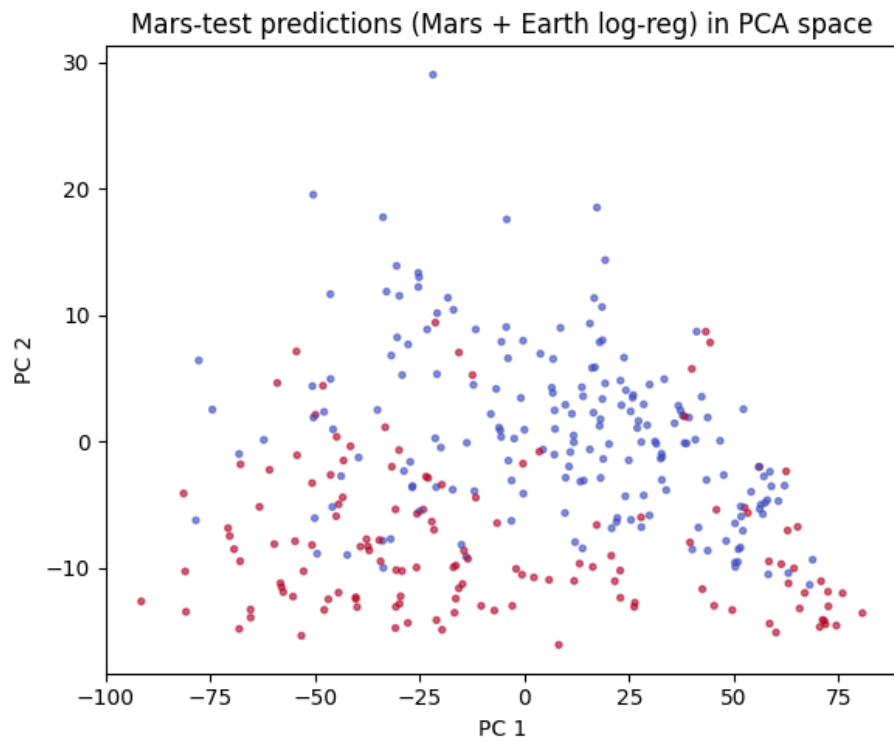


Figure 4 - PCA scatter of Mars-test coloured by Mars+Earth predictions

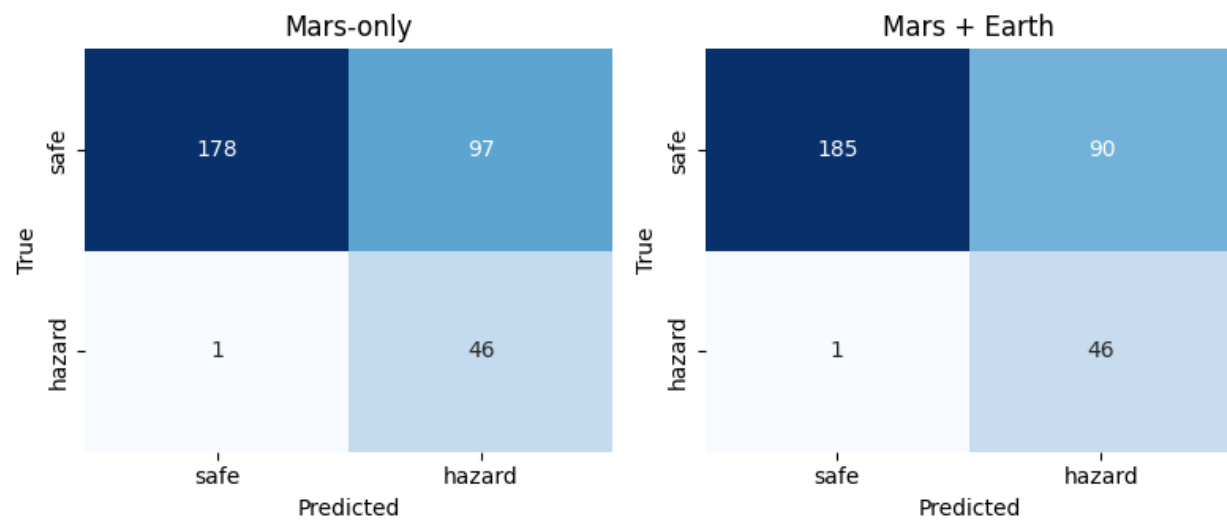


Figure 5 - Side-by-side confusion matrices